



**Cambridge Instruments  
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · HAND-DRIVEN, BATTERY-FREE

# CIE/444

Hand-cranked analog insulation tester in anti-magnetic aluminium body — no batteries required. Available in single-range configurations from 100V to 5000V.

BATTERY-FREE

## Insulation Tester — Metal Body, Hand-Driven

A hand-driven generator-type insulation tester for direct-reading insulation resistance on machines, switchgear, cables and transformers — power is generated entirely by the operator, no batteries or external supply required.

- **No batteries required** — hand-cranked generator
- **Centrifugal governor** — holds output voltage constant
- **Anti-magnetic aluminium body** — die-cast, rugged
- **Cross-coil indicator** — anti-parallax mirror scale
- **3-second response** — quick direct reading

Supplied as base instrument only — carrying case, test leads, and range options available at extra cost.

*Manufactured in India since 1963 — trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors.*



- V 100-5000V** Available ranges, specify at order
- 0 Battery-Free** Hand-cranked generator
- 3 3 Seconds** Response time

### 01 Full Specification Sheet

| PARAMETER                                     | SPECIFICATION                                                               |
|-----------------------------------------------|-----------------------------------------------------------------------------|
| <b>Body</b>                                   | Anti-magnetic aluminium                                                     |
| <b>Generator Type</b>                         | Hand-driven generator, centrifugal governor                                 |
| <b>Scale Length</b>                           | 90 mm (approx.)                                                             |
| <b>Accuracy</b>                               | Conforms to IS: 2992-1987                                                   |
| <b>Rotation Speed</b>                         | 160 r.p.m. or more                                                          |
| <b>High Voltage Test</b>                      | Min. 2000V or (rated V + 1000)V A.C. r.m.s., 1 min                          |
| <b>Insulation Resistance Test</b>             | Better than 50 M $\Omega$ at 500V D.C.                                      |
| <b>Guard Terminal &amp; Infinity Adjuster</b> | Provided on instruments with insulation range 1000 M $\Omega$ and above     |
| <b>Response Time</b>                          | 3 seconds                                                                   |
| <b>Rated Voltages Available</b>               | 100V / 250V / 500V / 1000V / 2500V / 5000V (single range per unit)          |
| <b>100V Resistance Range</b>                  | 0-20 · 0-50 · 0-100 M $\Omega$                                              |
| <b>250V Resistance Range</b>                  | 0-50 · 0-100 · 0-250 M $\Omega$                                             |
| <b>500V Resistance Range</b>                  | 0-100 · 0-200 · 0-500 · 0-1000 · 0-2000 M $\Omega$                          |
| <b>1000V Resistance Range</b>                 | 0-100 · 0-200 · 0-500 · 0-1000 · 0-2000 M $\Omega$                          |
| <b>2500V Resistance Range</b>                 | 0-1000 · 0-2000 · 0-2500 · 0-5000 · 0-10000 · 0-20000 · 50-50000 M $\Omega$ |
| <b>5000V Resistance Range</b>                 | 0-5000 · 0-10000 · 0-20000 · 50-50000 M $\Omega$                            |
| <b>Dimensions (L×W×H)</b>                     | 110 × 128 × 195 mm (approx.)                                                |
| <b>Weight</b>                                 | 2.5 kg (approx.)                                                            |
| <b>Warranty</b>                               | 12 months against manufacturing defects                                     |

*Any range apart from the standard voltage & resistance ranges above can be built to order.*

### 02 Typical Applications

- Motors, generators, and other rotating machinery
- Insulation control and switching equipment checks
- Distribution network and cable insulation testing
- Transformer, insulator, and installation checks
- Locating shorts, defects, and insulation failures

### 03 Optional Accessories

- + Carrying case
- + Test leads, 1m/2m
- + H.V. test leads, 2m
- + Dual voltage/range option

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reliability since 1963.